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Host Adaptation and Association with Mobile Genetic Elements in livestock-associated MRSA

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Staphylococcus aureus is an important human bacterial pathogen that has a wide host range, including livestock, companion, and wild animal species. Genomic and epidemiological studies show that S. aureus has jumped between host species many times over its evolutionary history.

Methicillin-resistant *Staphylococcus aureus* (MRSA) is a major drug-resistant pathogen responsible for a wide variety of infections worldwide, ranging from mild skin infections to life-threatening invasive diseases. Since its discovery, MRSA clonal complex 398 (CC398) has become a rapidly emerging cause of human infections, most often associated with livestock exposure.

The evolutionary relationships between hosts are a key factor in determining the success of a pathogen following transmission to a novel host species. *Staphylococcus aureus* sequence type (ST398) is a lineage affecting both humans and livestock worldwide. In this study, we sampled from and then assembled publicly available whole-genome sequence data from ST398. Our collection includes high-quality whole genome assemblies of 431 isolates spanning 19 host species (including humans, pigs, cows, chickens, turkeys, and horses, etc.), 43 countries (across Europe, North & South America, Asia, and Australasia), and 25 years (1999–2024).

We used this dataset and constructed a recombination-stripped maximum likelihood phylogeny of ST398's evolution. By comparing the genomes of isolates from livestock- and human-associated ST398, we concluded that the emergence of ST398 in livestock was associated with both the acquisition of antibiotic-resistance genes and the loss of genes associated with human immune evasion. This study confirms that the genes whose presence most strongly distinguishes isolates from the human- and livestock-associated groups are associated with three categories of MGEs: Tn916 transposon carrying tetM, (2) SCCmec carrying mecA, and (3) φSa3 prophages carrying a human immune evasion gene cluster. We further analyzed the SCCmec and φSa3 diversity and its adaptation to specific hosts and geographical locations within the LA clade.